

ASOPE and the fabled promise of process knowledge capture and reuse

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This paper addresses the question of how process knowledge should be structured. The use of aspect orientation as an abstraction for capturing and using process knowledge is reported. A description of the method, ASOPE (aspect oriented process engineering), is given, and the context of its development from ideas generated by research on software engineering is described. ASOPE introduces two new abstractions — generic process patterns and process aspects. An illustrative example of the use of these ideas is given. Finally, the limitations of the work are described and a number of avenues for further development are proposed.

1. Introduction

Business process knowledge is the knowledge that managers and consultants use to design and construct a business process. It often remains locked in the minds of such experts, or implicit in the structure of an organisation. In recent years, the value of representing this knowledge explicitly has been recognised; however, capturing such knowledge is a skilled and labour-intensive task. Reusing process knowledge derived in one context in another would save much time and effort. It would also enable best practice to be propagated within an organisation. The representation proposed here facilitates reuse by separating the description of the general structure of a family of processes from knowledge of how to specialise a generic description for use in a particular circumstance.

The ideas that are presented in this paper draw on research in software engineering concerning the way in which program modules should be encapsulated. Computer programs and business processes share a number of similarities in terms of their basic structures and properties. For example, they are both sequences of tightly defined, inter-dependent operations, and they both tend toward unmanageable complexity. It seems natural, therefore to draw the work of one community of investigators into the work of the other. This work has led to the development of a technology that has been dubbed ASOPE (aspect oriented process engineering).

ASOPE decomposes processes into two types of element. The first of these, generic process patterns (GPPs), are the sequences of fundamental steps that are used to achieve a class of business goal. The second, process aspects, contain the process steps that specialise the GPPs to

a specific context, together with instructions for their composition. Process aspects are context-specific process fragments that represent operational procedures such as safety checks. By combining process aspects and generic process plans it is possible to use ASOPE to synthesise wholly new processes. It should be noted that the process aspects allow the contextualisation to take place in a consistent way throughout a process. For example, a process aspect might introduce extra or alternative process steps that achieve sub-goals from different parts of a process in a complementary way. Another could implement a constraint to prevent the system from reaching a particular state. An example of a GPP would be a plan to move house. Process aspect examples would be safety checks such as checking for ponds on the premises in a process aspect related to an individual with children, or checking that the site had the appropriate access features for an elderly couple. GPPs are decontextualised, generalised plans, and process aspects are context, resource, and process participant dependent, plan exceptions and extensions.

The primary objective of decomposing process knowledge in this way is to construct a generative representation that facilitates the reuse of process knowledge in the form of successful plans developed in the past, and ‘must do’ elements derived from local working practice or policy decisions. A secondary objective of the work is to develop a representation that can be structured in a way that reflects the real structure of the organisation, which generates and maintains the knowledge. The objective of this paper is to describe a method of encapsulating and organising knowledge that might provide a suitable abstraction for this task.

1.1 Relationship to other work

ASOPE is an attempt to address the question of how process knowledge should be structured. This question is described as the 'key intellectual challenge' of developing tools and systems that can be used for the management of business processes [1].

The process handbook system developed by Malone's group at MIT [2] is probably the research effort closest in spirit to ASOPE. Several of the ideas that underpin the ASOPE architecture were drawn from the work on the process handbook. For example, specialisation of processes, an object-oriented knowledge structure and the view of processes as containing valuable knowledge that should be managed to facilitate organisational learning were all drawn from the work on the process handbook.

However, a sharp distinction exists between the process handbook and ASOPE. The use of process aspects enables the development of a process design meta-process that weaves information on the current context of the business process into its structure at run time. Also the use of a cross cutting¹ abstraction like aspect-orientation has an impact on the taxonomy of process objects that an ASOPE-style system implements. The idea of process aspects was derived from the techniques of software composition developed by the object-oriented programming community, specifically from the ideas of aspect orientation developed by Kiczales [3]. An aspect, as described by Kiczales, is a specialised program module which contains instructions that are to be 'woven' with the instructions in the modules that make up the main program. The idea is that the aspects are used to code elements of the program that are dependent on the context of the program's execution, and instructions to the 'weaver' (the 'aspect weaver' can be thought of as an interpreter) for their integration with the rest of the program.

There is a considerable literature describing methods that allow business processes to be modelled. Perhaps the most significant of these is the IDEF methodology [4] which was initially developed in response to the requirements of the US airforce for modelling the functional perspective of a system. The IDEF3 language describes processes in terms of 'units of behaviour' (UOB) and object state transition diagrams. An IDEF3 process flow description captures a description of a process and the network of relations between processes in the context of the scenario in which they occur to show how the process functions in a particular organisation. This is in contrast to the approach of untangling the business process from its context which is used in ASOPE. A more complete

description of IDEF can be found in de Witte and Pourteau [5].

Another group investigating process construction is the AI planning community. Research on AI planning focuses on systems that can construct chains or sequences of plan steps which, when executed, will result in some goal being achieved. For example, if a computer is to capture a piece in a chess match it must construct a plan of the sequence of moves to be executed in order to capture the piece. These plans are often found by searching the space of all possible plans and finding the 'best' possible way to achieve the goal. Plans are derived by either forward chaining from the current conditions of the domain to the goal conditions, backward chaining by finding a sequence of steps that will achieve the goal by working backwards from the desired state to the current state of the system, or by refinement of generic plans. The HTN (hierarchical task network) family of planning algorithms refine a partially ordered task sequence in an iterative, layered manner in much the same way as this paper advocates. Exemplars of HTN planners are O-plan [6] and UMCP [7]. The novelty of the work discussed in this paper lies in the use of an aspect-oriented representation of process knowledge, and the potential that this approach offers for process knowledge to be captured and reused by the organisation.

Workers in the AI-Planning community have also turned their attention to the importance of describing the domain on which the planner is working. The common process language [8] is an attempt to construct a process modelling language for general use in planning and process domains. CPL is based on first-order logic and describes a process in terms of the activities and objects that are executed and executed over. An important notion in CPL is that of a constraint, which represents some non-goal requirement of the process. In ASOPE, constraints on the process are not modelled explicitly; instead the approach is to utilise the knowledge stored in the process aspects to tacitly constrain the process. Our intuition is that by composing processes from elements that have been constructed by experts to conform to the constraints that they know must be met in order for the process to be executed successfully, the composed process will conform to the same constraints.

1.2 Structure of this paper

The structure of the paper is as follows. Section 2 describes the approach to storing and reusing process knowledge — ASOPE. In section 3 an illustrative example of how this technology can be applied to business processes is given. Finally in section 4 a summary of the work is presented together with how it is intended to extend and continue to develop this theme.

¹ That is affecting several different parts of the structure in complementary ways.

2. Aspect-oriented process engineering (ASOPE)

This section describes why it is appropriate to apply the ideas and principles that underlie aspect oriented programming to the domain of business process engineering.

Business processes are considered to be human-centric systems, and as such should be flexible and dynamic. However, business processes remain cumbersome and difficult both to construct and comprehend. This is because of the numerous 'special cases', exceptions and decision points that must be included in any realistic business process implemented by a large organisation that sets the needs of its customers as a priority. For example, a process developed to provide international private circuits for customers of BT contains 16 decision points on the top-level process, 16 top-level process tasks, and each of the top-level processes contain numerous decision points and exceptions themselves. Research techniques, such as the use of software agents, solve numerous problems in the provisioning and enactment stages of a business process [9—11], but research on process enactment systems such as workflow and agent-workflow has not considered the flexible creation of processes given the needs of the participants (consumer, employee, regulator, etc).

ASOPE considers the construction of specialised business processes from generic cores and context-dependent elements. New process steps or different process sequences can be implemented as new policies and instructions are disseminated across a business domain. The core essence of the process is abstracted from the detail of customer preferences, local working agreements and management instructions and then reused for different customers, workplaces and management domains.

In order to abstract the core of the business process from the context or customer-dependent elements that must be considered, the business process is defined in terms of two sets of components:

- generic process patterns — partially ordered processes, which consist of process steps, which are obligatory for a particular goal to be achieved (these are an ordered list of the conditions that must be satisfied for the goal to be successful),
- process aspects — process steps that can be included in a business process in order to customise it for execution by a particular resource.

A generic process pattern is the core process that must be executed in order to achieve a type of business goal. These are the processes that are publicly advocated for use by management and subscribed to by the operational units. The process aspects are the context-dependent process steps that should be carried out in order to achieve a particular

goal given a particular team, in a particular place for a particular customer.

By weaving process aspects that carry information on the process steps to be executed and information on how they should be incorporated, a specialised process can be derived (see Fig 1). The term 'weaving' is used to convey the idea that an aspect is a thread linking different parts of the process, specialising different tasks in consistent ways. Aspects are merged with the GPP according to weaving rules instantiated by the aspect weaver and the weave instructions that are coded into the aspect. The process aspects used are a subset of the overall set of aspects and the business process context determines their use.

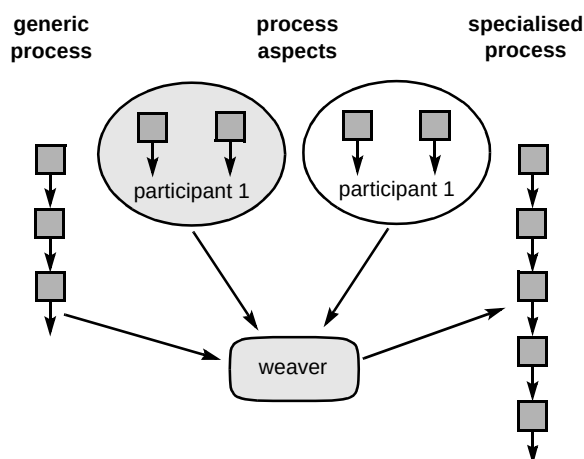


Fig 1 Process participants weave their particular steps into the generic process to create a specialised ordered process for execution.

During each weave a process aspect can customise the process in two ways:

- process addition (introduce weave) — a process step is added either before or after a previously defined step,
- process redefinition (advise weave) — a process step is redefined to make it relevant to the business process context.

An advise weave inserts new process instructions into a particular step in the process. The process position at which an aspect's advise weave is to be woven is determined by the weaver matching to templates defined in the aspect. These patterns take the form of a match to a process step name with wild cards matching to the preconditions and post-conditions of the process step. Because the processes are arranged hierarchically using a frame-based object-oriented organisation, this kind of match and weave process can be used to merge process sequences together. The use of an object-oriented organisation of the business process definitions was also chosen to promote the reuse of process definitions and enforce consistency within a process hierarchy.

The assumption underlying the ASOPE approach is that the participants in the meta process of constructing and managing the enactment of the business process belong to particular management domains which are controlled by managers with limited, but real and valuable, autonomy. This enables the participants to determine which resources will be utilised to achieve a goal on the basis of the impact that their use will have on the characteristics of the process. Figure 2 shows the process elements specialised by an ordinate layer being passed to a sub-ordinate to incorporate its process knowledge.

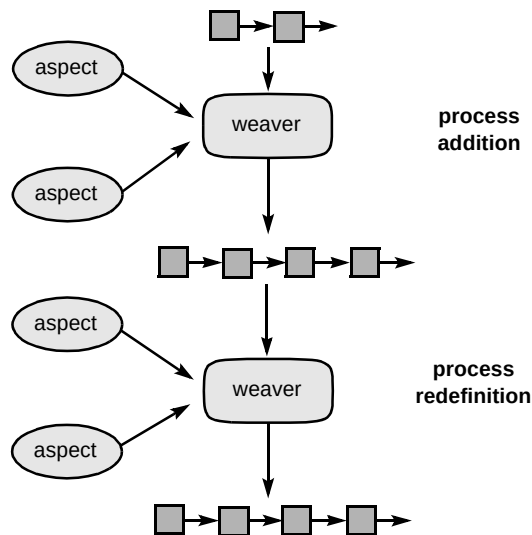


Fig 2 The hierarchical, iterative specialisation of a business process as it passes through management domains.

This arrangement results in the incremental specialisation of the process as it passes through management domains. At each level of the management domain the process is specialised by the weaver using the process aspects stored for that domain.

The next section describes an example of how this method of organizing and managing process knowledge can be applied to an everyday business activity.

3. Customised processes for telephone installation

Many commercial organisations have a core activity that concerns the installation or servicing of equipment at a customer's site or domicile — for example, gas, water and electricity utilities must install, service and read meters in order to bill their customers.

Consider the example of installing a phone, a business process that has some significance for BT. A simple set of core generic process steps is required for all customers, no matter where they are in the country, or which engineer does the installation. However, some steps will be different from customer to customer. A corporate customer may require adherence to a previously agreed contract, breach of

which may be unacceptable. Providing a service in a rural area may involve more travel time, but it will be unnecessary to book a parking bay for the engineer's van; however, in the City of London parking space may need to be pre-arranged.

In order to apply ASOPE to the process of installing a telephone it is necessary to define the participants and the generic process of installation to be used. For the purposes of this example two different (simplified) customer types are considered:

- customer type 1 — customer's time is paramount,
- customer type 2 — customer's security is paramount.

While the goal of installing a telephone is identical for both of these customer types, they are qualitatively very different. Customer type 1 will place a premium on appointments being kept, and probably can supply a mobile phone number that allows them to be contacted a short time before the technician arrives to carry out the work. Customer type 2 will be available for the visit of the technician and will require a special identification procedure to reassure them that the technician is genuine.

At present a single process is used to achieve the goal of installation for both groups of customer. However, by localising the knowledge of the extra process steps each customer requires in a process aspect, the process can be untangled to provide a customised process for both customer types. Suppose other aspects are developed which code specialisations for other customer types. If these are selected then specialised processes for these particular customers can also be derived.

A demonstration system has been implemented that uses the ideas described in this paper to specialise process definitions by weaving them with additional aspect-based information. The implementation built uses the Aspect-JTM aspect weaver to compose process aspects and generic process plans of the sort described in section 2 into new, specialised process definitions. An aspect for each customer type to be considered is defined. An aspect hierarchy could be created that can be used to inherit behaviour, but for this example the two customer aspects will be considered in isolation (see Fig 3). The process aspects shown in Fig 3 contain information on process steps to be taken, coded in this case as simple print statements, and weave information coded in the form of 'advise' weaves. These 'advise' instructions instruct the Aspect-JTM weaver to place the weaves before or after process steps in the generic plan pattern — coded as the method `planPattern` in this case.

```

aspect CustomerType1 extends Customer{
  int customerPhoneNumber = 0;
  advise
    *installPhone.bookVisit(*) {
      after {
        System.out.println("Acquire customers
        mobile contact number");
        customerPhoneNumber = 12345678;
      }}

  advise
    *installPhone.installEquipment(*) {
before { System.out.println("Verify
  appointment with customer number = " +
  String.valueOf
  (thisAspect.customerPhoneNumber)); }}}

aspect CustomerType2 extends Customer{
  advise *installPhone.installEquipment
  (*){

    before {System.out.println("Verify
    code word with customer"); }}

  advise * installPhone.bookVisit(*){

    after {System.out.println("Supply
    code word to customer"); }}

```

Fig 3 Customer specific process steps coded as Aspect-J™ aspect instructions.

Although the representation used is unsuited to a planning domain, the example serves to demonstrate how the process can be made to work in practice. A generic process pattern is defined to be used as the basis for the process (see Fig 3). The aspects are instantiated, as is the process pattern, in an Aspect-J™ module. Each customer's process aspect is added to the process pattern in turn to produce customised processes (see Fig 4).

```

public void acquireJobDetails() {

  System.out.println("Acquire
  appropriate job information");}

public void bookVisit () {
  System.out.println("Book visit time
  and address, allocate engineer") ; ; }

public void waitUntilVisit () {
  System.out.println ("Wait for visit
  time"); }

public void installEquipment () {
  System.out.println ("Carry out
  appropriate installation work");}

public void billCustomer() {
  System.out.println("Send bill to customer");}

public void planPattern () {
  acquireJobDetails();
  bookVisit();
  waitUntilVisit();
  installEquipment();
  billCustomer(); }

```

Fig 4 A generic process pattern for phone installation coded as an Aspect-J™ object.

Even the simple example above shows the need for aspect instances in this application. Customer type 2 requires that a telephone number is set for each appointment; it is appropriate to associate this information with the customer, and an aspect instance is an appropriate way of handling this (see Fig 5).

process for customer type 1 is:

 acquire appropriate job information

 book visit time and address, allocate engineer,
 acquire customer's mobile contact number

 wait for visit time

 verify appointment with customer; phone number = 12345678

process for customer type 2 is:

 acquire appropriate job information

 book visit time and address, allocate engineer

 supply code word to customer

 wait for visit time

 verify code word with customer

Fig 5 Customised phone installation processes generated by weaving the process aspects in Fig 3 with the GPP in Fig 4. Note the difference in the instruction sequence for these processes, but their common core structure.

4. Conclusions

This section of the paper summarises the work, along with any limitations, and looks at the future directions for research.

4.1 Summary

This paper has introduced a new approach to expressing and reusing process knowledge. Knowledge is expressed as generic process patterns and process aspects. These are woven together using the information specified in the weave instructions encoded in the process aspects. This makes it possible to remove the context-dependent elements of a business process and reuse its core sequence of operations. Removing the context-dependent steps of the process enables us to reapply those steps to other processes that are to be instantiated within the same context.

There are three business benefits to be gained from using this new technology:

- firstly, the core business process that underlies any particular activity can be articulated and subscribed to

and reused across all the management domains of the organisation,

- secondly, local practices can be captured and propagated as best practice by the organisation and/or retained for reuse in the future within a particular management domain,
- thirdly, processes can be tailored to the interests of those that have some stake in achieving the process goal, potentially improving both customer satisfaction and employee motivation as well as increasing organisational flexibility.

4.2 Future work

It is apparent that there are a number of limitations within the current work on ASOPE.

Firstly, it is clear that there are deficiencies in the current generation of aspect weaving tools. Aspect-JTM was used as the aspect weaving engine for the work in this paper by coding the process aspects and generic process patterns as Aspect-JTM code and using the program that was generated by the weaver to output the plan. As it is currently implemented Aspect-JTM uses random weave orderings and has no facilities for specifying aspect precedence. However, AOP is a nascent field and as it evolves it is likely that features such as rules for weave ordering will become incorporated into the tool sets, thus modelling the complexity of the real world more closely. It is worth noting that investigators in the field of artificial intelligence have tackled these issues in a number of different ways when working on logic programming, blackboard architectures and rule chaining. Our use of the Aspect-JTM weaver was a pragmatic decision which enabled us to investigate the potential of using aspect orientation for process engineering. In the longer term it will be necessary to either develop our own weavers, or alternatively, if Aspect-JTM becomes more suited to our purpose, we will have to wrap it in some way. It is also clear that a substitution, or filtering weave is necessary in order for the aspect-oriented abstraction to be really useful both for process representation and software development. Composition filters have been used to develop aspect-oriented systems that directly implement deletion mechanisms which are not implemented in Aspect-JTM [12] and we can see no obstacle to the implementation of more weave instructions. Currently the appropriate, sufficient and necessary weave instruction set for process engineering remains unclear and is at this time an open research issue.

As mentioned above the prototype emits processes in the form of an informal text description. An obvious extension is to emit process descriptions in a format (e.g. PIF [13] or IDEF) that allows them to be loaded directly into current process management tools such as workflow management systems. This enables rapid generation of new

customised processes, but would not allow the processes to be modified subsequently as information that needs to be stored within the aspect (for example the customer number used in the example in this paper) would be lost. We are currently investigating how we can overcome this.

Another clear limitation is the conceptual difficulty involved in the design of the weave orderings, and the correct decomposition and encapsulation of the process elements into generic process patterns and process aspects. This is a problem that has long been the bane of any object-oriented or frame-based representation. There are no hard rules for the development of object models, simply best practices and rules of thumb which apply only to the particular circumstances of the program and programmer who is structuring a particular model. Unfortunately it is clear that while aspect orientation offers another tool which can be used to simplify the organisation of knowledge, the success of the development of a knowledge base will continue to depend on the skills and judgement of the knowledge engineer doing the development.

One approach to dealing with the difficulties of designing and constructing processes using ASOPE would be to present the information to the users of the system in an intuitive manner. We intend to explore ways that ASOPE can present process aspects and GPPs to users in diagrams, and also explore the generation of process aspects and GPPs from other process definitions, possibly in a dialog with the user.

ASOPE is work in progress; to date primitive tools have been developed that take hand-crafted process aspects and generic process patterns coded in Aspect-JTM, and weave them into various processes.

The next step in the research will be to provide graphical support to these activities so we can continue the investigation without spending half of our time debugging the code.

In the medium term the vision is to develop tools that will allow the build, maintenance and use of process repositories of the type developed for the process handbook. It is at this stage that it will become clear if ASOPE offers any real advantage, or whether it is just an interesting exercise in structuring process knowledge in different ways.

In the even longer term the development of specialised weavers for process engineering may be required. Currently we are attempting to reuse the 'off-the-shelf' weavers developed by the AOP community, in particular Aspect-JTM. In the longer term, issues of performance may mean that we have to develop bespoke weavers that support only process engineering as the core of a truly aspect-oriented process knowledge repository and process engineering environment.

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